

EduStrat: A Browser-Based Tool for Teaching Quantitative Analysis of Educational Inequality with PISA Microdata

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Software

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Summary

EduStrat (Educational Stratification in PISA) is a browser-based, open-source tool that enables students, instructors, and researchers to explore how parental socioeconomic characteristics predict student academic achievement using microdata from the OECD Programme for International Student Assessment [PISA; OECD (2024)]. The tool covers eight PISA cycles (2000–2022), more than 100 countries, and approximately 3.5 million student observations. Users select countries and years through an interactive interface; the application loads pre-generated data subsets on demand and performs all statistical computations client-side—requiring no software installation, no server, and no programming. EduStrat computes survey-weighted descriptive statistics, inequality indices (Gini coefficient, Lorenz curves, percentile ratios), regression models (pooled OLS, fixed effects, random effects) with diagnostic tests, achievement gap decomposition, and variance decomposition. Where the data carry PISA’s replicate weights, design-correct standard errors are computed by balanced repeated replication (BRR, Fay’s method) rather than the simple-random-sampling formulae. Every estimator is verified numerically against independent R reference implementations (see *Verification and reproducibility*). All outputs can be exported as CSV tables, publication-quality figures (PNG/SVG), or self-contained HTML reports with embedded analytic metadata. In accordance with the OECD Terms of Use for PISA Public Use Files, the application does not redistribute micro-level student data; all exports provide computed estimates and aggregated statistics only.

Statement of Need

Courses in comparative education, educational policy, and quantitative social science frequently engage with PISA data to illustrate concepts such as educational inequality, intergenerational transmission, and cross-national variation in achievement. However, working with PISA microdata presents substantial pedagogical barriers. The data arise from complex stratified sampling designs requiring proper survey weighting; achievement is released as plausible values; and results are sensitive to choices about weights, country selections, and regression specifications (Mislevy et al., 1992; OECD, 2009; Wu, 2005). Teaching these concepts typically requires students to write code in R, Stata, or Python before producing even basic descriptive summaries—diverting class time from substantive learning to software troubleshooting.

Existing tools partially address this gap. The learningtower R package (Wang et al., 2024) provides cleaned PISA extracts, but requires R proficiency. The OECD’s PISA Data Explorer offers a web interface but is limited to pre-tabulated summaries without

42 regression modelling or diagnostic testing. The *intsvy* (Caro & Biecek, 2017) and *EdSurvey*
43 (Bailey et al., 2024) R packages provide comprehensive analytical capabilities but assume
44 substantial programming experience.

45 EduStrat fills the space between these tools: it provides the analytical depth of R-based
46 packages (regression, decomposition, diagnostics) in an accessible browser interface that
47 requires no installation or coding. This allows instructors to focus class time on interpreting
48 results, comparing specifications, and understanding methodological choices—rather than
49 debugging software environments. Students can immediately engage with questions like:
50 *How does the ESCS gradient differ between Finland and the United States? How does*
51 *the Hausman test inform the choice between fixed and random effects? What happens to*
52 *inequality metrics when we change the country selection?*

53 **Functionality and Teaching Use**

54 EduStrat implements the core quantitative methods taught in graduate courses on
55 educational inequality and comparative education, organised across interactive analysis
56 tabs.

57 **Survey-weighted descriptive statistics.** Students learn why sampling weights matter
58 by toggling between weighted and unweighted results and observing how estimates change.
59 The tool computes weighted means, standard deviations, and quantiles following OECD
60 guidelines (OECD, 2009).

61 **Inequality metrics.** The Gini coefficient, coefficient of variation, percentile ratios
62 (P90/P10), and Lorenz curves allow students to compare distributional properties of
63 achievement across countries and understand what each metric captures.

64 **ESCS gradients and quartile gaps.** The central measure of intergenerational
65 transmission—the regression slope of achievement on the ESCS index (Avvisati & Wuyts,
66 2024)—is computed with interactive visualisations showing scatter plots and fitted lines.
67 Quartile-based gaps (Q4–Q1) with Cohen’s *d* provide intuitive effect size interpretation.

68 **Regression model comparison.** Students estimate pooled OLS, country fixed-effects,
69 and random-effects models side by side, with coefficient tables, diagnostic plots (residual
70 vs. fitted, Q-Q plots), and the Hausman specification test. This teaches specification
71 sensitivity: how conclusions about the SES–achievement relationship change depending
72 on modelling choices. Standard errors default to BRR replicate-weight estimates when the
73 data provide them, and the interface labels which method is active—making visible how
74 much PISA’s complex sampling design inflates uncertainty relative to the naive formulae.

75 **Variance decomposition.** Within- and between-country decomposition with intraclass
76 correlation illustrates multilevel structure in educational data.

77 **Export system.** Results are exportable as CSV, PNG/SVG, or self-contained HTML
78 reports with embedded analytic metadata (countries, years, variables, weight choice, model
79 specification). This supports reproducibility and teaches students to document their
80 analytic decisions. Per OECD Terms of Use, exports provide aggregated statistics and
81 computed estimates only—not individual student records.

82 EduStrat was conceived as a visual, instructional explorer of how parental socioeconomic
83 status (the ESCS index) stratifies student achievement in PISA, and was developed
84 iteratively with feedback from colleagues and instructors at both upper-secondary and
85 tertiary levels, with the aim of improving its analytical functionality and its value for
86 learning. The application was presented to colleagues in the Department of Communication
87 and Public Policy at the Università della Svizzera italiana and to students in a bachelor’s-
88 level social-science statistics course in April 2026.

89 The intended contribution is accessibility: students can engage with intergenerational
 90 transmission, survey weighting, and specification sensitivity directly—without first learning
 91 R, Stata, or Python—so that class time can be spent interpreting results and weighing
 92 methodological choices rather than troubleshooting software. We make no claim of
 93 improved learning outcomes: the tool has not been evaluated in an experimental setting
 94 or a controlled classroom study. This is an acknowledged limitation of the present work,
 95 and a systematic evaluation of the tool’s effect on student learning could form part of a
 96 future expansion.

97 Project Story

98 EduStrat originated from the practical challenge of teaching quantitative methods
 99 in comparative education. Preparing PISA microdata for classroom demonstrations
 100 required writing substantial data-wrangling and analysis code before each session, and
 101 students without R or Stata experience could not independently replicate or extend the
 102 analyses shown in class. The tool was developed to package these recurring analytical
 103 operations—weighted descriptives, gradient estimation, specification comparison, variance
 104 decomposition—into a self-contained web application that students could use immediately.

105 The intellectual work behind EduStrat is not the production of JavaScript, HTML and
 106 CSS; it is the set of decisions about *which* methods belong in an instrument for teaching
 107 educational stratification, *how* each should be estimated so that a browser reproduces what
 108 R or Stata would produce, and *how* to make the methodological choices that PISA forces
 109 on every analyst visible to students rather than hidden inside a library. Those choices—the
 110 ESCS gradient as the operational measure of intergenerational transmission (Avvisati
 111 & Wuyts, 2024); the fixed- versus random-effects contrast as a lesson in specification
 112 sensitivity; survey weighting and replicate-weight variance estimation as the difference
 113 between description and inference—are the scholarship; the code is their implementation.

114 The application is static and client-side: all computation runs in the browser in JavaScript,
 115 with no server-side component, so it can be served from any static host or run locally.
 116 The codebase comprises 21 ES6 JavaScript modules (~10,400 lines) with documentation
 117 covering methodology, data sources, and a variable codebook. The data architecture
 118 reflects a key constraint—serving roughly 3.5 million student records to a browser: rather
 119 than loading the full ~1.25 GB dataset, EduStrat pre-generates 513 country-year JSON
 120 files and loads only the requested subsets, and a pipeline of R scripts extracts, chunks,
 121 and validates the microdata from the learningtower package (Wang et al., 2024). In
 122 accordance with the OECD Terms of Use, the repository redistributes no micro-level data;
 123 the hosted demonstration serves the generated chunks.

124 Two methodological points received particular attention because they bear directly on
 125 whether the tool teaches *inference* or only *description*. First, EduStrat uses a single
 126 plausible value per domain—a constraint inherited from learningtower, which distributes
 127 only PV1—so the measurement (imputation) component of variance is not estimated;
 128 this is stated in the interface and is itself used as a teaching point about plausible values.
 129 Second, and in response to the recommendation that PISA estimates use the OECD’s
 130 replicate weights, the project moved beyond learningtower (which ships only the final
 131 weight `W_FSTUWT`): a dedicated pipeline script re-sources the raw OECD Public Use Files
 132 and regenerates chunks that carry the 80 Fay replicate weights, so that standard errors
 133 can be computed by balanced repeated replication exactly as the OECD prescribes. This
 134 was implemented and verified across three cycles (2015, 2018, 2022) for a set of countries
 135 as a documented, reproducible template rather than a full re-release. The design-correct
 136 standard errors are noticeably larger than the naive ones, by a factor that is stable across
 137 cycles (for the Finnish mathematics mean, the BRR error is 1.7–2.0 times the simple-
 138 random-sampling error in 2015, 2018 and 2022 alike)—making the cost of ignoring the
 139 sampling design something students can see, not merely be told.

Verification and reproducibility

Because EduStrat performs all computation client-side in JavaScript, the central scholarly question is whether those in-browser estimates can be trusted. The project answers this with a verification harness that treats the JavaScript artifact as untrusted and holds it to established statistical software. The harness runs the application's *own* analysis modules in Node—providing only the same two numeric libraries the browser loads—against real PISA chunks, and compares every reported quantity against an independent reference computed in R from peer-reviewed packages: `stats::lm` for weighted OLS and least-squares dummy-variable fixed effects, `plm` (Croissant & Millo, 2008) for the Swamy–Arora random-effects estimator and the Hausman test, `lmtest` (Zeileis & Hothorn, 2002) for the Breusch–Pagan test, `car` (Fox & Weisberg, 2019) for variance inflation factors, `stats::cooks.distance` for influence, and `intsvy` (Caro & Biecek, 2017) for BRR replicate-weight standard errors.

Across 83 computational checks (point estimates, standard errors, test statistics, R^2 , degrees of freedom) and 21 replicate-weight checks spanning three cycles, all agree with the R references; the large majority match to between ten and fourteen significant figures. A headless-browser smoke-test additionally confirms that the application runs in a real browser engine with no errors and that the rendered output reports the BRR standard errors. The few quantities that do not match to machine precision are documented honestly rather than tuned away: the random-effects slope matches `plm` to seven significant figures, and the Hausman statistic—which divides the squared fixed-/random-effects contrast by a near-zero variance difference—is reproduced to about two parts in a thousand, a sensitivity inherent to the statistic rather than an error in its implementation. Several estimators were corrected during this process (the weighted Gini coefficient, the Breusch–Pagan test, the variance inflation factors, the Cook's distance leverages, and the Hausman variance term), with the harness serving as the regression test. The full procedure, tolerances, and per-method results are recorded in the repository (VERIFICATION.md, pipeline/scripts/04-verify-computations.R, pipeline/scripts/06-verify-brr.R) and can be re-run by any reviewer.

Use of generative AI

Generative AI tools (Claude, Anthropic) were used to write the application code (JavaScript, HTML, CSS) and to help draft documentation. The author, an educational sociologist, directed the design and methodological choices, specified the estimators and their intended behaviour, and—critically—supervised all AI-generated computation through the R-based verification described above: no statistical method is reported as correct on the strength of its having been generated, only on the strength of its agreement with an independent R reference that the author wrote and can re-run. The verification scripts, not the generated code, are the warrant for the results. Remaining limitations (single plausible value; limited-scope replicate-weight coverage) are documented in the interface and in this paper.

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